Eigenvalues and inverse problems

Rank-two maximizers of spectral spread on an entry interval

Difficulty: challenging **Importance:** interesting to the community

Status: open in general; several dimension and interval cases proved.

Problem statement

For an integer $n \geq 2$ and $a \in [-1, 1)$, let

$$\mathcal{S}_n[a, 1] = \{A \in \mathbb{R}^{n\times n} : A = A^T,\ a \leq A_{ij} \leq 1 \text{ for all } i, j\}.$$

The spread of A is $s(A) = \lambda_{\max}(A) - \lambda_{\min}(A)$. The Fallat–Xing conjecture asserts that there exists a matrix B of rank exactly two, every entry of which belongs to $\{a, 1\}$, such that

$$s(B) = \max_{A\in\mathcal{S}_n[a,1]} s(A).$$

All diagonal entries are free to vary within the same interval as the off-diagonal entries. The assertion concerns existence of a rank-two maximizer; it does not assert that every maximizer has rank two. The normalization $[a, 1]$, $-1 \leq a < 1$, is the source's reduction of the arbitrary nondegenerate real-interval problem. All dimensions and intervals form one problem.

Why it matters

The conjecture reduces an eigenvalue optimization over a full symmetric matrix family to a low-rank extremizer. It sharpens spectral bounds obtainable from entrywise uncertainty alone.

References

S. M. Fallat and Y. Xing, *On the spread of certain normal matrices*, Linear and Multilinear Algebra 60 (2012), 1391–1407, §2. N. J. Calkin, R. M. Corless, L. Gonzalez-Vega, J. R. Sendra, and J. Sendra, *On the maximal spread of symmetric Bohemian matrices*, 2025, §1 (displayed Fallat–Xing conjecture), §§3.2, 7–8, and §10.

Status check

Version 1 of the 2025 paper is the latest listed version checked. It proves the conjecture for all $a \in (-1, 1)$ when $2 \leq n \leq 7$, and for $a = 0$ when $2 \leq n \leq 8$ or 3 divides n. The case $a = -1$ was already proved by Zhan. Its conclusion explicitly records a gap in the authors' attempted general proof. The authors' supporting repository confirms the finite computational ranges. Searches for *Fallat Xing spread conjecture proof 2026* and later maximal-spread papers found no general resolution. Restricting entries to interval endpoints alone is already known and is not counted separately.